

H

Roll No. 229703J

TMA-316

B. TECH. (THIRD SEMESTER)

END SEMESTER

EXAMINATION, Dec., 2023

**DISCRETE MATHEMATICAL STRUCTURES
AND COMBINATORICS**

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Define transitive closure of relation. Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 3), (3, 1)\}$. Find the transitive closure of R using Warshall's algorithm. (CO1)

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- (b) Define a Lattice. Give an example of an infinite lattice with : (CO1)
- (i) neither a least nor a greatest element.
 - (ii) a least but not a greatest element.
 - (iii) a greatest but not a least element.
 - (iv) both a least and a greatest element.
- (c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$, where $\mathbb{R}$ is the set of real numbers. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2$ and $g(x) = x + 4$. State whether functions are injective, surjective or bijective. (CO1)
2. (a) In a bolt factory, three machines M_1 , M_2 , and M_3 manufacture 2000, 2500, and 4000 bolts every day. Of their output 3%, 4%, and 2.5% are defective bolts. One of the bolts is drawn very randomly from a day's production and is found to be defective. What is the probability that it was produced by machine M_2 ? (CO2)

- (b) A die is thrown 6 times. If “getting an odd number” is a “success”. What is the probability of : (CO2)

- (i) 5 successes
- (ii) at least 5 successes
- (iii) at most 5 successes

- (c) A random variable has the following probability distribution : (CO2)

x	$P(x)$
4	0.1
5	0.3
6	0.4
8	0.2

Find the expectation and Standard Deviation of random variable.

3. (a) Define a valid argument. Check the validity of the following argument : If I take breakfast, then I go to school. I do not take breakfast. Therefore, I do not go to school. (CO3)

- (b) Explain quantifiers. Symbolize the following sentence using (i) universe of discourse and (ii) without universe of discourse : (CO3)

(1) All states in India are not highly populated.

(2) Some students are tall but not healthy.

- (c) Using mathematical induction prove that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n . (CO3)

4. (a) Consider the set Q of rational numbers, and let $*$ be the operation on Q defined by $a * b = ab/2$. Check whether $(Q, *)$ is an abelian group ? (CO4)

- (b) Find the solution of the recurrence relation : (CO4)

$$a_n + 2a_{n-1} - 3a_{n-2} = 2^n$$

- (c) Define a cyclic group. If G is a cyclic group, then prove that G has exactly two generators. (CO4)

5. (a) Define a connected graph. A disconnected simple graph G (without self-loops and parallel edges) with n vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges. (CO5)
- (b) Prove that the minimum height of a full binary tree with n vertices is $\lceil \log_2(n+1) - 1 \rceil$, where $\lceil x \rceil$ is the smallest integer greater than or equal to x . (CO5)
- (c) Define Euler and Hamiltonian Graphs and give the example of graph : (CO5)
- (i) Euler but not Hamiltonian
 - (ii) Hamiltonian but not Euler
 - (iii) Both Euler and Hamiltonian